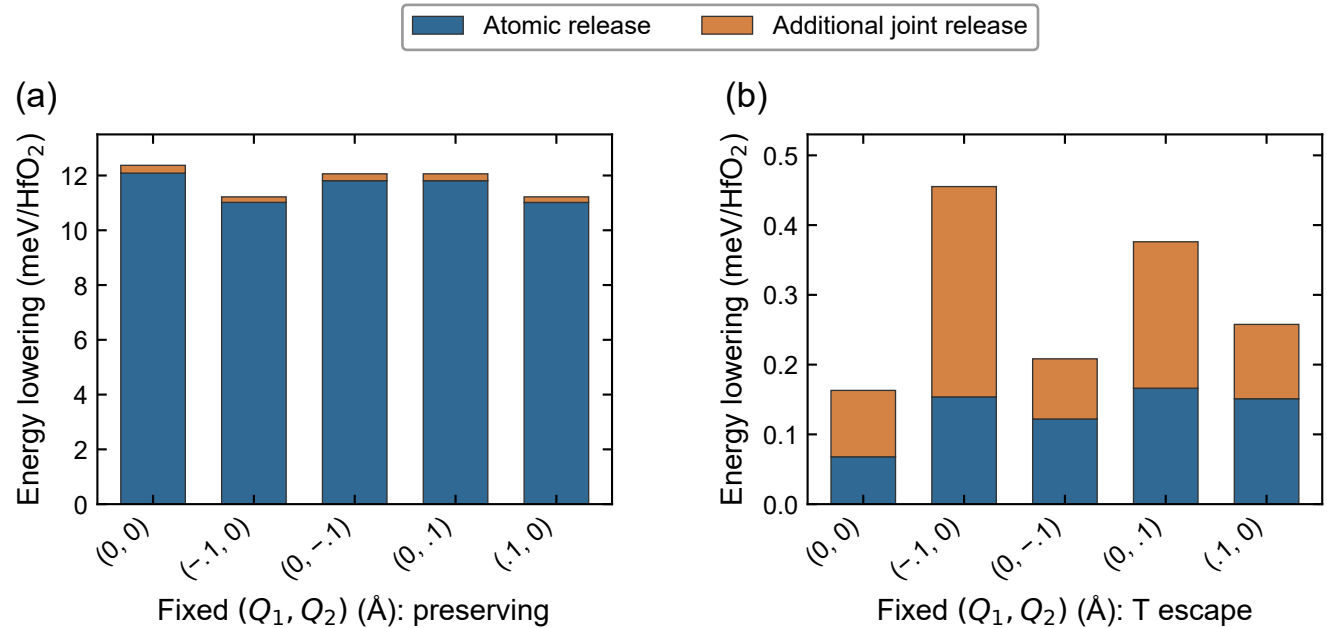
Atomic release
Additional joint release
(a)
Energy lowering (meV/HfO2)
0
2
4
6
8
10
12
(0, 0)
(−.1, 0)
(0, −.1)
(0, .1)
(.1, 0)
Fixed (Q1, Q2) (Å): preserving
(b)
Energy lowering (meV/HfO2)
0.0
0.1
0.2
0.3
0.4
0.5
(0, 0)
(−.1, 0)
(0, −.1)
(0, .1)
(.1, 0)
Fixed (Q1, Q2) (Å): T escape